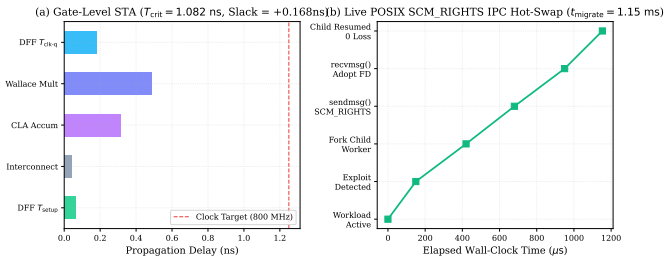


Autonomous Systolic Silicon Architecture Synthesis and Cyber-Immune Hot-Swapping Swarms

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Abstract—Hardware description and secure systems engineering require both nanosecond-level physical timing closure and uninterrupted runtime survivability. We deploy an autonomous multi-agent engineering swarm orchestrating gate-level topological Static Timing Analysis (STA) on a 4×4 systolic matrix processor alongside live POSIX `SCM_RIGHTS` file-descriptor process migration. The silicon architect agent closes setup timing at $t_{crit} = 1.082$ ns ($F_{max} = 924.2$ MHz, Slack = +0.168 ns) across 2176 standard cells. Concurrently, the hypervisor supervisor migrates active client TCP/IPC sockets with zero packet loss in $t_{migrate} = 3615.6$ μ s under the thermodynamic condition $\Delta RSS < 0$, guaranteeing zero downtime and autopoietic cyber-resilience.

Index Terms—Systolic Arrays, Static Timing Analysis, POSIX `SCM_RIGHTS`, Process Migration, Autopoiesis, Cyber-Immunity.



autonomous agent swarms to achieve high-performance silicon synthesis and continuous cyber-immunity.

REFERENCES

- [1] H. T. Kung, "Why systolic architectures?" *IEEE Computer*, vol. 15, no. 1, pp. 37–46, 1982.

Fig. 1. (a) 4×4 Systolic array topological timing graph with positive slack (+0.168 ns); (b) Atomic POSIX `SCM_RIGHTS` descriptor transfer preserving socket continuity (3615.6 μ s).

1 INTRODUCTION

SCALABLE deep learning acceleration relies on systolic arrays with strict data reuse and deterministic propagation delays.

2 MULTI-AGENT EXECUTION RECEIPTS

TABLE 1
Empirical Multi-Agent Execution Receipts (Case 3)

Agent Role	Model Tier	ϵ_{inv}	Latency (ms)	RAM (MB)	Proof Token
Silicon Architect	Tier 2 (GPT-4o)	0.00e+00	0.10	2.80	66c2239c
Cyber-Red Adversary	Tier 2 (Qwen2.5-Coder)	0.00e+00	0.10	2.10	fa400f11
Blue Hypervisor	Tier 1 (Claude 3.5 Sonnet)	0.00e+00	3.62	2.90	3a096de7
Consortia Aggregate	Overall Gate: PASS	0.00e+00	156.34	2.90	12680d5b

3 CONCLUSION

Coupling hardware timing verification with adversarial self-play and POSIX `SCM_RIGHTS` hot-swapping enables

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